

Multiple Choice (10 marks)

1. Why are Cepheid stars relevant for astronomers?
 - (a) To measure interstellar mass.
 - (b) To measure galactic distances.
 - (c) To measure galactic energy-density.
 - (d) To measure interstellar density.
2. What is the Cassini division of Saturn's rings?
 - (a) a dark ring visible from Earth composed of dark dusty particles
 - (b) the most opaque ring of Saturn made of highly reflective ice particles
 - (c) the widest ring of Saturn located between two large ring gaps
 - (d) a large gap visible from Earth produced by an orbital resonance with the moon Mimas
3. The sky is blue because
 - (a) the Sun mainly emits blue light.
 - (b) the atmosphere absorbs mostly blue light.
 - (c) molecules scatter red light more effectively than blue light.
 - (d) molecules scatter blue light more effectively than red light.
4. The astronomical unit parsec (pc) plays a crucial role in astronomy. One parsec is equal to about 3.26 light-years. How is one parsec defined in astronomy?
 - (a) Distance at which one astronomical unit measures one arcsecond from Earth.
 - (b) Orbital distance of the solar system around the center of the Milky Way in one year.
 - (c) Effective distance of the solar wind (i.e. the radius of the heliosphere).
 - (d) Historical distance to the brightest star Sirius.
5. Approximately how old is the surface of Venus?
 - (a) 750 million years.
 - (b) 2 billion years.
 - (c) 3 billion years.
 - (d) 4.5 billion years.

True / False (10 marks)

Answer True or False

6. True or False: Asteroids emit a significant amount of their own radiation, which contributes to their visibility in the night sky.
7. True or False: Ganymede, being the largest moon in the solar system, is also the closest moon to Jupiter among its four major satellites.
8. True or False: On a planet with half the gravity of Earth, a 5 kg sand bag would weigh 50 N, indicating no change in weight despite reduced gravity.
9. True or False: All fresh impact craters on solid surfaces typically exhibit ejecta and raised rims as common features.
10. True or False: Dione is not one of the Galilean moons, which include Ganymede, Callisto, Europa, and Io.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of Pluto's orbital period of approximately 250 years on its classification as a dwarf planet and its influence on our understanding of the Kuiper Belt.
12. Discuss the role of asteroids and comets in the context of the Solar Nebular theory, focusing on their origins as leftover planetesimals and their significance in planetary formation.

End of Paper